

## Practice Final: Answer Key

### Problem 1 (15 points)

- (a) **Yes.** Gauss's Law is a fundamental mathematical consequence of the  $1/r^2$  nature of the Coulomb force (via the Divergence Theorem). It is absolutely true for *any* closed surface, regardless of the symmetry or the materials involved.
- (b) **No.** To pull  $\vec{E}$  out of the integral ( $\oint \vec{E} \cdot d\vec{a} = E \oint da$ ), the electric field must have constant magnitude and a consistent dot product with the area vector everywhere on the surface. An irregularly shaped chunk of metal breaks all spherical symmetry, making the integral analytically unsolvable by hand.
- (c)  $\vec{E} = 0$ . In electrostatics, the electric field inside the bulk material of any conductor must be exactly zero. If it were not, free charge carriers would feel a force and move until the field was neutralized.

### Problem 2 (20 points)

- (a) The total resistance is the sum of the series components:  $R_{\text{tot}} = \rho_1 \frac{L_1}{A} + \rho_2 \frac{L_2}{A}$ .  
The power dissipated is  $P = I^2 R_{\text{tot}} = I^2 \left( \frac{\rho_1 L_1 + \rho_2 L_2}{A} \right)$ .
- (b)  $\nabla \cdot \vec{J} = 0$  **everywhere**. By the continuity equation ( $\nabla \cdot \vec{J} = -\partial\rho/\partial t$ ), a steady current means charge is not accumulating anywhere over time ( $\partial\rho/\partial t = 0$ ). Therefore, the divergence of  $\vec{J}$  is identically zero everywhere, including at the interface.
- (c)  $\nabla \cdot \vec{E} \neq 0$  **at the interface** between the two materials.  
*Sign: Positive.* Because  $J$  is continuous and uniform, we use Ohm's Law ( $\vec{E} = \rho\vec{J}$ ). Since  $\rho_2 > \rho_1$ , the electric field in the second region must be stronger ( $E_2 > E_1$ ). By Gauss's Law, a jump in the electric field requires a surface charge. Since the field increases in the direction of the area vector, the divergence is positive, indicating a static layer of *positive* surface charge  $q_s$  has built up at the interface.

### Problem 3 (25 points)

- (a) **Yes, it experiences a net force to the right.** Because  $+q$  is off-center, it induces a non-uniform distribution of  $-q$  on the inner cavity wall. The negative charge concentrates heavily on the wall closest to  $+q$  (the right wall). This asymmetric induced charge exerts a net attractive force on  $+q$ , pulling it further to the right.
- (b) **No.** As established in (a), the charge crowds the wall closest to the  $+q$  charge. By Gauss's Law, a surface just inside the metal must enclose zero net charge, so the total integrated charge on the inner wall is exactly  $-q$ .
- (c) **Yes.** The interior charges ( $+q$  and the induced  $-q$ ) produce an electric field of exactly zero inside the bulk of the conductor. Therefore, no information about the asymmetry of the cavity charge leaks to the outside. By the Uniqueness Theorem, the  $+q$  on the outer surface distributes itself to maintain an equipotential surface. Without any internal field acting on it, it distributes perfectly uniformly, completely blind to the fact that the inner charge is off-center.
- (d) **No.** The metallic shell acts as a perfect electrostatic shield. Grounding the outer surface simply allows the outer  $+q$  charge to flow to earth, eliminating  $\sigma_b$  and the external field. However, the field inside the bulk of the metal remains zero. Therefore, the boundary condition for the inner cavity is perfectly unchanged, and the force on  $+q$  remains exactly the same.

#### Problem 4 (20 points)

- (a) **Increase.** The charges  $+Q$  and  $-Q$  attract each other. Pulling them apart requires an external agent to do positive mechanical work against this attractive Coulomb force. This work is stored as additional potential energy in the electric field.
- (b) *Energy Method:*  $C_i = \frac{\epsilon_0 A}{s}$  and  $C_f = \frac{\epsilon_0 A}{2s}$ . Because it is isolated,  $Q$  is constant.  

$$\Delta U = U_f - U_i = \frac{Q^2}{2C_f} - \frac{Q^2}{2C_i} = \frac{Q^2(2s)}{2\epsilon_0 A} - \frac{Q^2 s}{2\epsilon_0 A} = \frac{Q^2 s}{2\epsilon_0 A}.$$
*Work Method:* The field from one plate is  $E_1 = \frac{Q}{2A\epsilon_0}$ . The force on the other plate is  $F = QE_1 = \frac{Q^2}{2A\epsilon_0}$ . The work done to move the plate a distance  $s$  is  $W = F \times s = \frac{Q^2 s}{2\epsilon_0 A}$ . The two exactly match.

- (c) **Push further apart.** The electroscope measures the potential difference  $V$ . Since  $V = Q/C$  and  $C$  has halved (while  $Q$  remained constant), the potential  $V$  has doubled. A higher potential will cause the leaves to deflect more.

**Problem 5 (20 points)**

- (a) To ensure  $V = 0$  at  $z = 0$ , every real charge requires an opposite-sign image charge mirrored across the  $xy$ -plane.

- Image 1:  $-q$  at  $(0, d, -d)$  (mirrors the real  $+q$ )
- Image 2:  $+q$  at  $(0, -d, -d)$  (mirrors the real  $-q$ )

- (b) The total work is the energy required to assemble the real charges in the presence of the plane.

**Step 1:** Bring in real  $+q$ . Work is done against its own image ( $-q$ ).

$$W_1 = \frac{1}{4\pi\epsilon_0} \frac{q(-q)}{2d}$$

**Step 2:** Bring in real  $-q$ . Work is done against the real  $+q$ , the image of  $+q$ , and its own image ( $+q$ ).

$$W_2 = \frac{1}{4\pi\epsilon_0} \left[ \frac{(-q)(+q)}{2d} + \frac{(-q)(-q)}{\sqrt{(2d)^2 + (2d)^2}} + \frac{(-q)(+q)}{2d} \right]$$

$$\textbf{Total Work: } W_{\text{tot}} = W_1 + W_2 = \frac{q^2}{4\pi\epsilon_0} \left[ -\frac{1}{2d} - \frac{1}{2d} + \frac{1}{d\sqrt{8}} - \frac{1}{2d} \right] = \frac{q^2}{4\pi\epsilon_0 d} \left( \frac{1}{\sqrt{8}} - \frac{3}{2} \right)$$

## Answer Key: Circuits

### Problem 6

- (a)  $I = \mathcal{E}/R_1$ . At  $t = 0^+$ , uncharged capacitors resist sudden changes in voltage, meaning  $V_C = 0$ . They behave perfectly as ideal, zero-resistance wires (shorts). Because  $C_2$  acts as a zero-resistance wire, it perfectly shorts out resistor  $R_3$ . The current bypasses  $R_3$  entirely.  $C_1$  also acts as a short, putting  $R_2$  in parallel with the  $C_2$  short. The entire right side of the circuit has 0 equivalent resistance, leaving only  $R_1$  to impede the battery.
- (b)  $I = \mathcal{E}/(R_1 + R_3)$ . As  $t \rightarrow \infty$ , the capacitors fully charge and act as open circuits (infinite resistance breaks in the wire).  $C_1$  blocks all current from flowing through the middle branch (so no current through  $R_2$ ).  $C_2$  blocks current from the far right branch. The only remaining unbroken path for DC current is the outer loop through  $R_1$  and  $R_3$ .
- (c)  $\mathbf{V}_{C2} = \mathcal{E} \frac{R_3}{R_1 + R_3}$  and  $\mathbf{V}_{C1} = \mathcal{E} \frac{R_3}{R_1 + R_3}$ .  
 For  $C_2$ , it is strictly in parallel with  $R_3$ , so it must share its voltage drop. By the voltage divider rule on the active outer loop,  $V_{R3} = IR_3 = \mathcal{E} \frac{R_3}{R_1 + R_3}$ .  
 For  $C_1$ , we apply KVL to the middle loop. Because no current flows through  $R_2$  in the steady state ( $I_2 = 0$ ), there is zero voltage drop across it ( $V_{R2} = 0$ ). Therefore, the entire potential difference between the top and bottom nodes must be dropped across  $C_1$ , which is exactly the same node-to-node potential difference as  $V_{R3}$ .
- (d) **False.** The student assumes all three resistors are in series. However, in the steady state, the branch containing  $R_2$  is an open circuit. Because  $I_2 = 0$ , no power is dissipated by  $R_2$  whatsoever ( $P_2 = 0^2 R_2 = 0$ ). The correct power dissipation is  $P = \mathcal{E}^2/(R_1 + R_3)$ .

### Problem 7

- (a) In a DC steady state, a fully charged capacitor acts as an open circuit, meaning **absolutely zero current flows through the middle branch**. Therefore, the complex multi-loop network collapses into a single, simple outer loop. A  $3 \times 3$  matrix is unnecessary because the junction rule at Node A simplifies to  $I_{\text{left}} = I_{\text{right}}$ .

- (b)  $I = \frac{\mathcal{E}_1 - \mathcal{E}_2}{R_1 + R_2}$ . Since the middle branch is "dead" to DC current, we trace the outer loop. The batteries oppose each other, so the net EMF is  $\mathcal{E}_1 - \mathcal{E}_2$ . The total series resistance of the outer loop is  $R_1 + R_2$ .

- (c)  $Q = C \left[ \frac{\mathcal{E}_1 R_2 + \mathcal{E}_2 R_1}{R_1 + R_2} \right]$ . (Top plate is positive).

To find  $Q$ , we need the potential difference between Node A and Node B. Let's set Node B as our ground reference ( $V_B = 0$ ).

We can find  $V_A$  by walking up the left branch from ground:  $V_A = \mathcal{E}_1 - IR_1$ .

Substitute the current from part (b):  $V_A = \mathcal{E}_1 - \left( \frac{\mathcal{E}_1 - \mathcal{E}_2}{R_1 + R_2} \right) R_1$ .

Find a common denominator:  $V_A = \frac{\mathcal{E}_1 R_1 + \mathcal{E}_1 R_2 - \mathcal{E}_1 R_1 + \mathcal{E}_2 R_1}{R_1 + R_2} = \frac{\mathcal{E}_1 R_2 + \mathcal{E}_2 R_1}{R_1 + R_2}$ .

Because there is zero current in the middle branch,  $R_3$  drops zero voltage. The entire potential difference  $V_A - V_B$  sits across the capacitor. Thus,  $Q = CV_A$ . Since  $\mathcal{E}_1, \mathcal{E}_2 > 0$ , Node A is at a higher potential than Node B, meaning the top plate is positive.

- (d) Let  $I_1$  be the current leaving  $\mathcal{E}_1$  (rightward),  $I_2$  be the current leaving  $\mathcal{E}_2$  (leftward), and  $I_3$  be the current flowing straight down the middle branch.

**KCL (Node A):**  $I_1 + I_2 - I_3 = 0$

**KVL (Left Loop):**  $\mathcal{E}_1 - I_1 R_1 - I_3 R_3 = 0$

**KVL (Right Loop):**  $\mathcal{E}_2 - I_2 R_2 - I_3 R_3 = 0$